MIL-C-22931/11A
20 January 1972
SUPERSEDING
MIL-C-22931/11
10 February 1968

MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, SEMIRIGID, COAXIAL, SEMI-AIR-DIELECTRIC, .875 TO 1.005 INCHES OUTSIDE DIAMETER, 50 OHMS

The complete requirements for procuring the cables described herein shall consist of this document and the latest issue of Specification MIL-C-22931.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

REQUIREMENTS:

Operating temperature range: -55 to +80° C.

Inner conductor: See table I.

Dielectrics:
Cable core and insulating layer: See table I.

Outer conductor: See table I.

Jacket: See table I.

Voltage withstand: 3,400 volts rms (minimum) at 60 Hz, or
6,000 volts dc.

Insulation defects, spark: See table I.

Attenuation: See table I.

Velocity: See table I.

Capacitance: See table I.

Impedance: 50 ±1 ohms.

Voltage standing wave ratio: See table I.

Mandrel size for cold bend, bending and temperature cycling tests: See table I.

Minimum bending radius: 10 inches (254.0 mm).

Part number: M22931/11- (dash number from table I).

FSC 6145

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TABLE I. Dash numbers and characteristics.

Requirement 1/	Dash number								
	001	002	003	004	005	006	007	008	009
Inner conductor	Copper wire or tubing		Copper wire		Copper tube			Copper covered aluminum wire	
Overall diameter (±0.003(.08))	0.300(7.62)		0.311(7.90)		0.358(9.09)		do.	.350(8.89)	Do.
Dielectrics									
Cable core	Polystyrene		Polyethylene		Polyethylene			Polyethylene	
Insulating layer	Polystyrene	do.	Not applicable	do.	Not applicable	do.		Not applicable	
Outer conductor	Aluminum tubing, smooth, seamless		Aluminum tubing, smooth, seamless		Copper tube, corrugated		Aluminum tube, corrugated	Copper tube, corrugated	
Inside diameter (nominal)	0.758(19.25)		0.833(21.16)		0.795(20.19)		0.795(20.19)	0.830(21.08)	
Outside diameter	0.875(22.23)		0.953 ±.003 (24.21) (.08)		1.005 ±.010 (25.53) (.25)		1.005 ±.010 (25.53) (.25)	0.980 ±.010 (24.89 ±.25)	
Jacket	Not applicable	Polyethylene	Polyethylene	Not applicable	Not applicable	Polyethylene		Polyethylene	Not applicable
Diameter		0.980(24.89) +.020(.51) -.010(.25)	1.093(27.76) +.037(.94) -.017(.43)			1.115 ±.030 (28.32) (.76)	do.	1.065 ±.020 (27.05) (.51)	
Minimum wall thickness		0.049(1.24)	0.049(1.24)			0.045(1.14)		0.050(1.27)	
Insulation defect, spark	Not applicable	Applicable	Applicable	Not applicable	Not applicable	Applicable		Applicable	Not applicable
Attenuation (dB/100 ft max)									
30 MHz	0.27		0.29		0.21		0.25	0.21	
400 MHz	1.04		1.03		0.9		0.95	0.9	
3,000 MHz	3.4		3.3		2.5		2.97	2.5	
Velocity (percent), nominal	90		85.5		91.6			94	
Capacitance (pF/ft)	22.5		24		22.2			21.0	
VSWR									
Initial (max)									
500-2,000 MHz	1.15	do.	1.15	do.	1.15	do.		1.15	Do.
2,000-5,000 MHz	1.15		1.15		1.30		do.	1.30	
After temperature cycling									
500-2,000 MHz	1.20		1.20		1.15			1.15	
2,000-5,000 MHz	1.20		1.20		1.30			1.30	
After bending									
500-2,000 MHz	1.20		1.20		1.25			1.25	
2,000-5,000 MHz	1.20		1.20		1.40			1.40	
Mandrel size radius	17.5(444.5)		10(254.0)		10(254.0)			10(254.0)	

1/ Dimensions are in inches; millimeters are in parentheses.
Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

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NOTES:

1. Cross reference information is as follows:

Part number	RG number	Former classification Class	Type
M22931/11-001	RG-197/U	N	I
M22931/11-002	RG-232/U		
M22931/11-003	RG-254/U		
M22931/11-004	RG-255/U		
M22931/11-005	RG-269A/U	N	II
M22931/11-006	RG-318/U		
M22931/11-007	None	None	None
M22931/11-008	None	None	None
M22931/11-009	None	None	None

2. Figures are for information.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU, MI
Navy - SH, EC
Air Force - 11, 17, 80
DSA - ES, IS

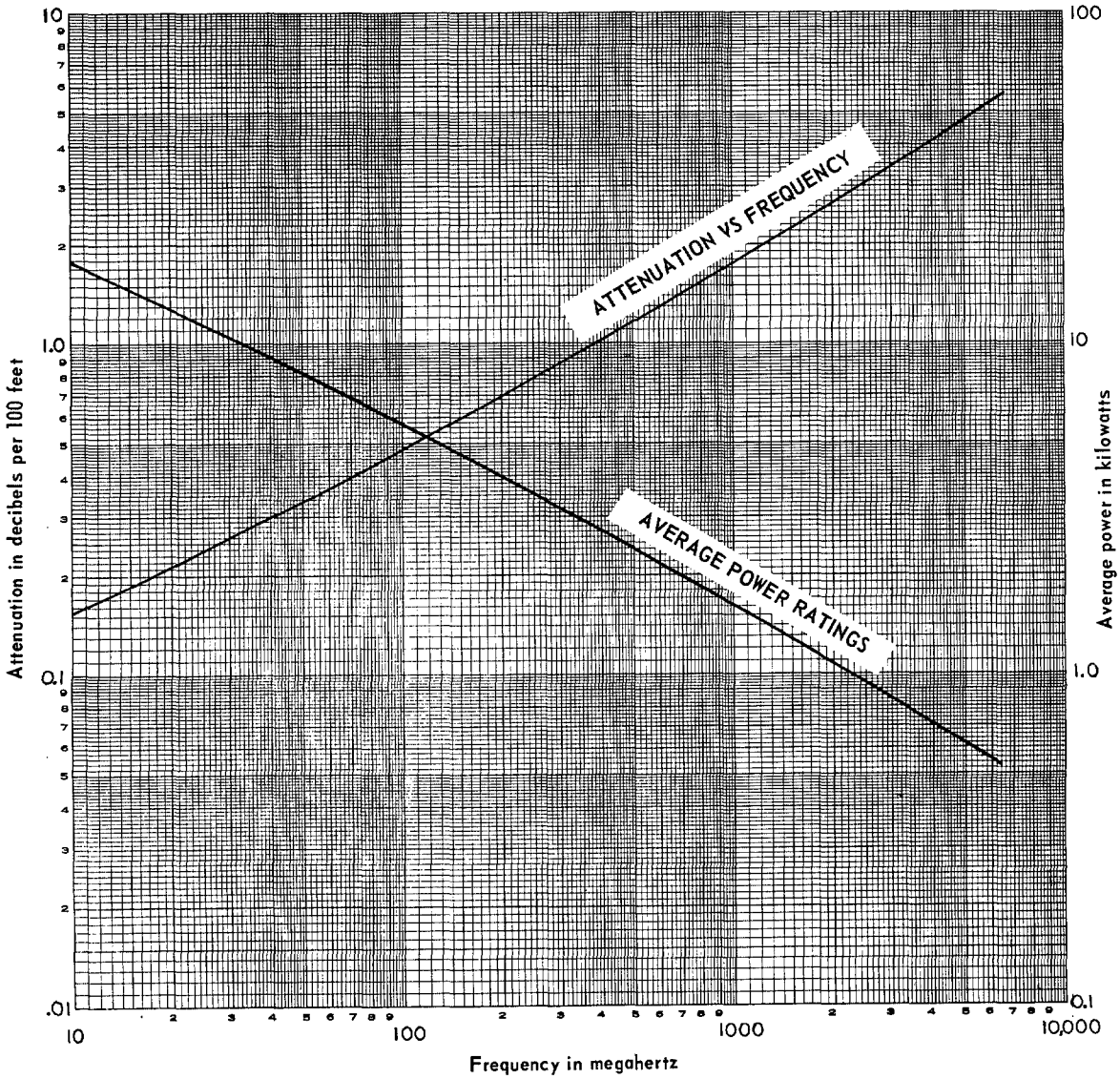
User activities:
Army - ME, AT, SL
Navy - AS, OS, MC
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 6145-0603)

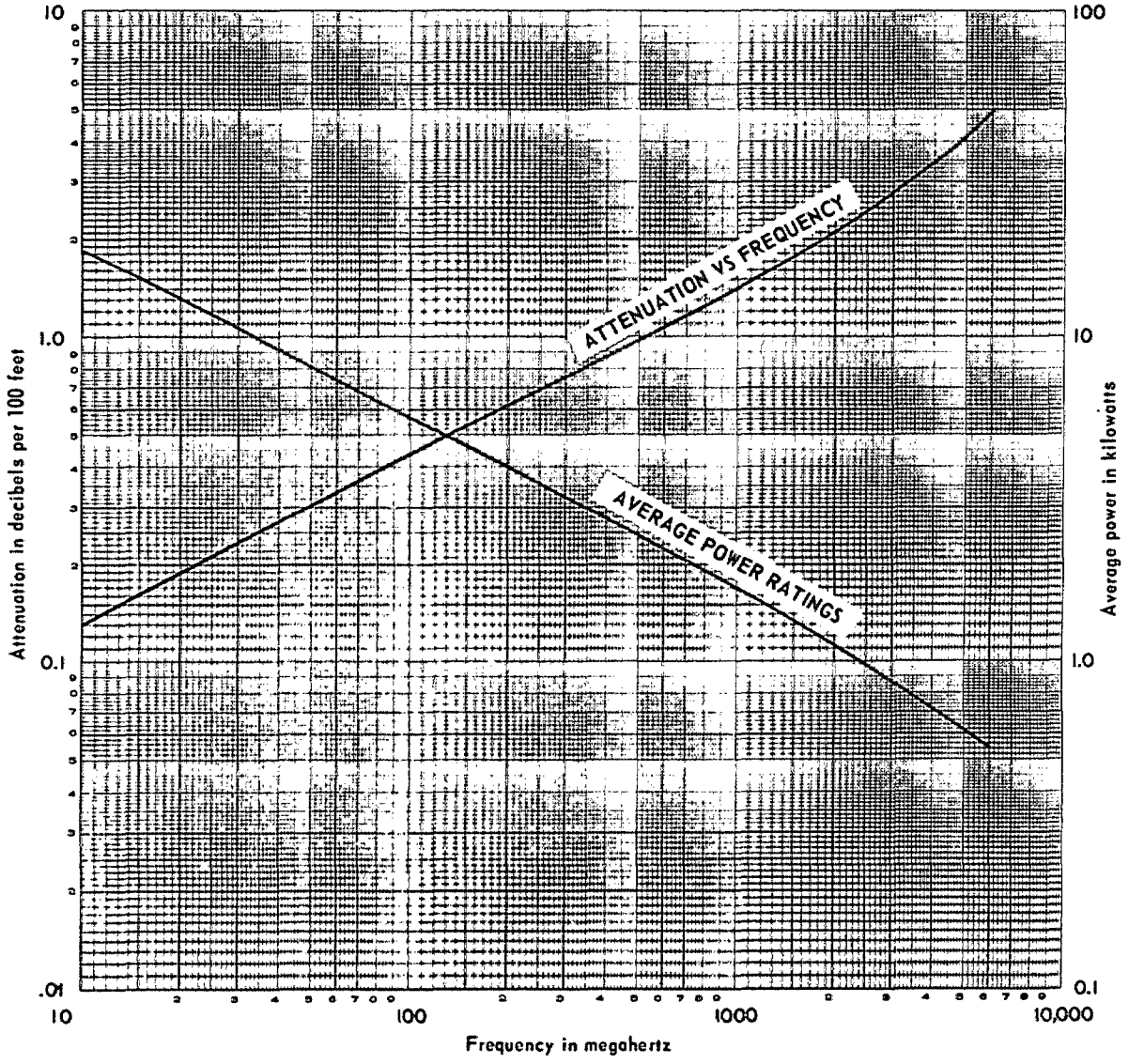
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Power rating based upon a 40°C ambient with a 45°C rise in center conductor temperature.

FIGURE 1. Attenuation and power rating curves for –001 and –002.

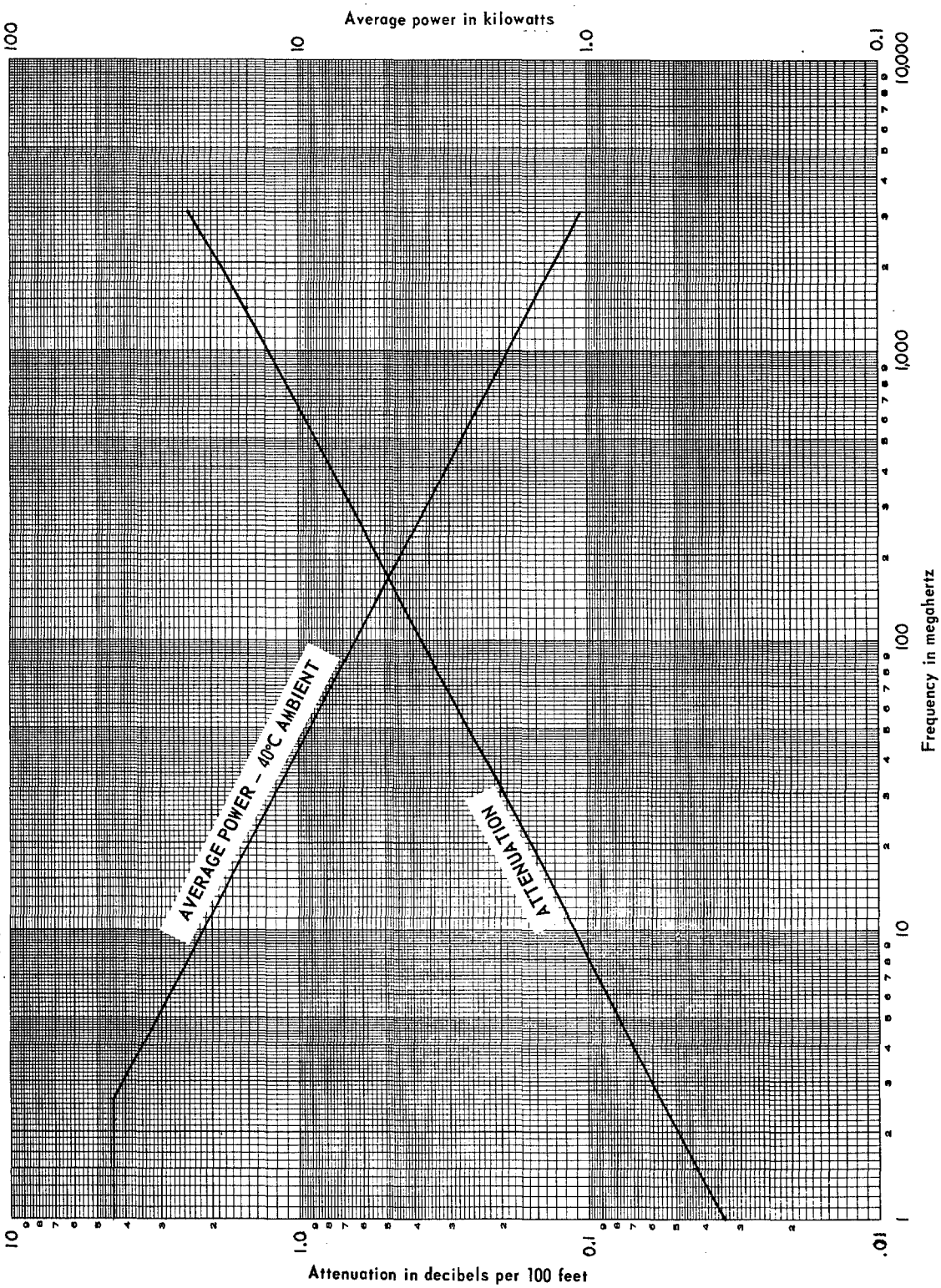
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Power rating based upon a 40°C ambient with a 45°C rise in center conductor temperature.

FIGURE 2. Attenuation and power rating curves for –003 and –004.

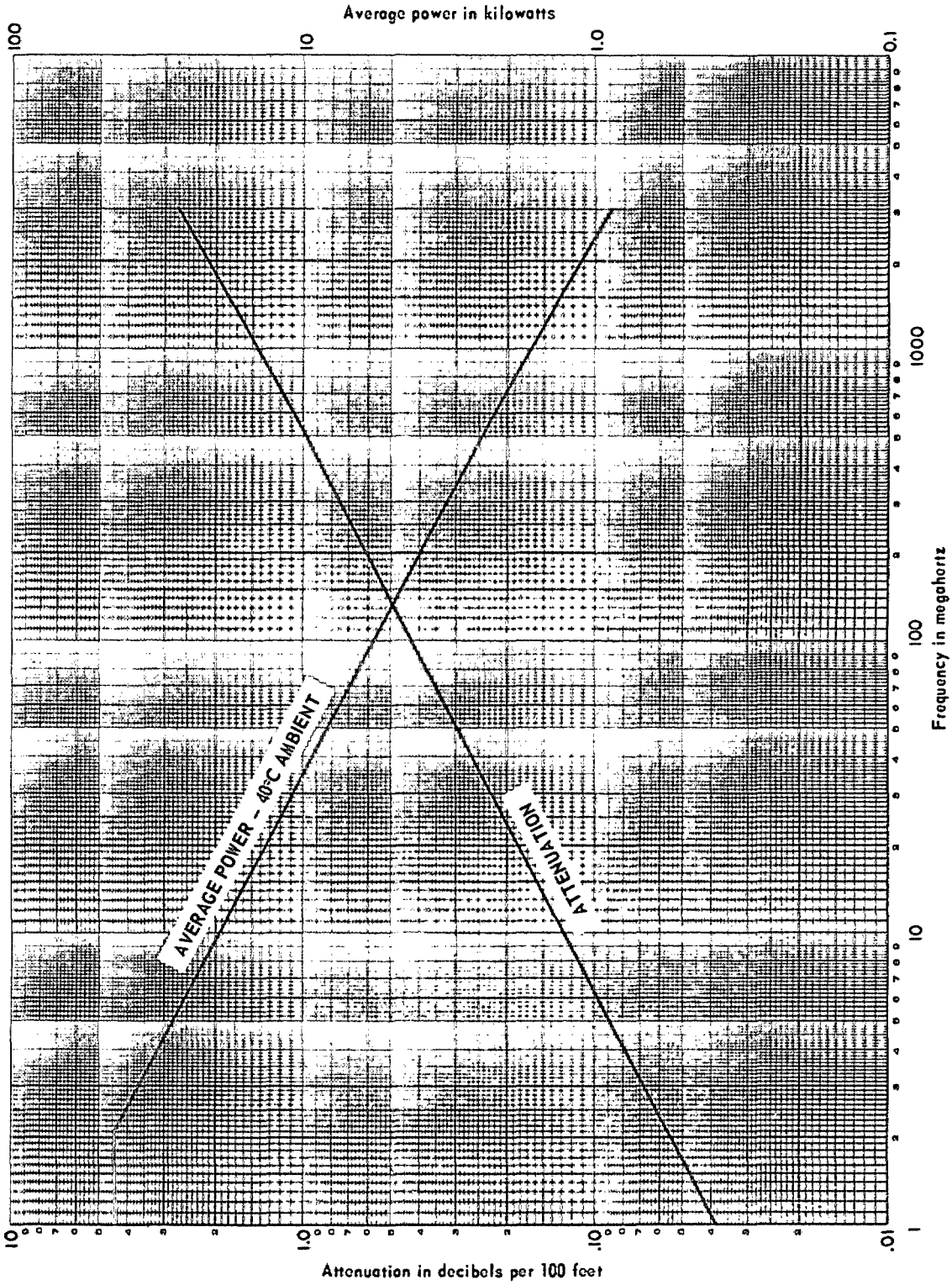
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Power rating based upon a 40°C ambient with a 45°C rise in center conductor temperature.

FIGURE 3. Attenuation and power rating curves for –005 and –006.

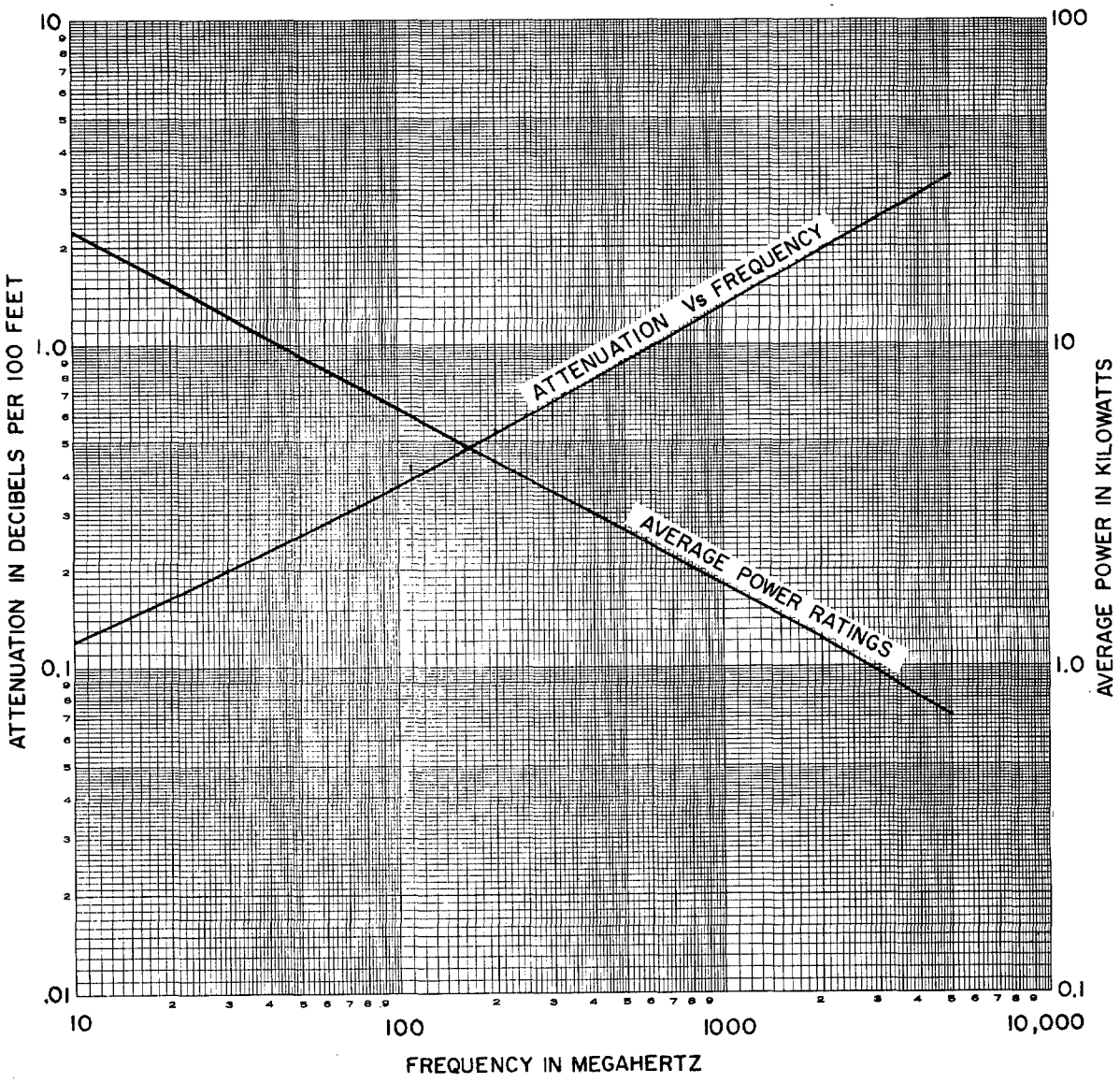
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Power rating based upon a 40°C ambient with a 45°C. rise in center conductor temperature.

FIGURE 4. Attenuation and power rating curves for –007.

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POWER RATINGS BASED ON:
VSWR 1.0
AMBIENT TEMPERATURE 40° C
INNER CONDUCTOR TEMPERATURE 80° C

FIGURE 5. Attenuation and power rating curves for 008 and 009.

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